

Graph-Guided Reference Screening for Diffusion-Generated Crystals

A proposal to improve Crys-JEPA's screening-and-refinement loop

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Motivation: Crys-JEPA [1], MatterGen [2], and MatterSim [3]

Research objective

Question

Can a **chemistry-constrained graph over generated candidates and known reference phases** rank stable crystals better than Crys-JEPA's equal-weight embedding-distance heuristic?

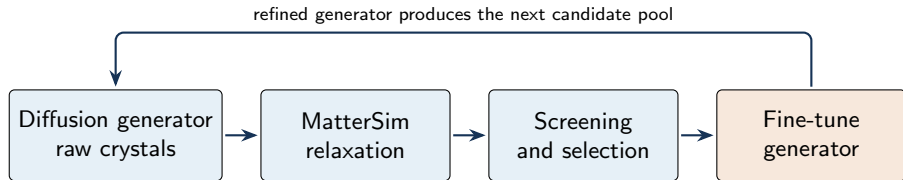
Why it matters

- Diffusion models can generate many plausible periodic structures.
- Only a small fraction will be stable and novel after validation.
- Better screening can improve the data used for generator refinement.

Contribution hypothesis

- Crys-JEPA retrieves chemically related phases but aggregates them uniformly.
- A GNN can learn which phase relationships are useful for ranking.
- The main experiment keeps the encoder frozen for clean attribution.

Context: generation, relaxation, and refinement



- **Relaxation** adjusts atomic coordinates and lattice parameters to reduce predicted energy and forces.
- MatterSim is a machine-learned force field: much cheaper than DFT, but not a substitute for final DFT validation [3].
- Our intervention is the **screening** box: improve what is selected for refinement.

What MatterGen contributes

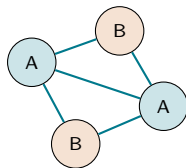
MatterGen is a diffusion model that jointly denoises:

- discrete atomic species;
- fractional coordinates on a periodic cell;
- lattice parameters.

It uses an equivariant, periodic atom-neighbour graph inside each crystal [2].

Important distinction

MatterGen's graph represents **atoms within one crystal**. This proposal builds a second graph whose nodes are **complete crystals**.



periodic atom graph
inside one candidate

Crys-JEPA screening and the proposed extension

Crys-JEPA learns an energy-aware crystal embedding:

$$\text{crystal } x \mapsto \mathbf{h}_x.$$

For a relaxed candidate c , it retrieves references $R(c)$ from the same chemical system and subsystems, and uses

$$D(c) = \frac{1}{|R(c)|} \sum_{r \in R(c)} \|\mathbf{h}_c - \mathbf{h}_r\|_2^2.$$

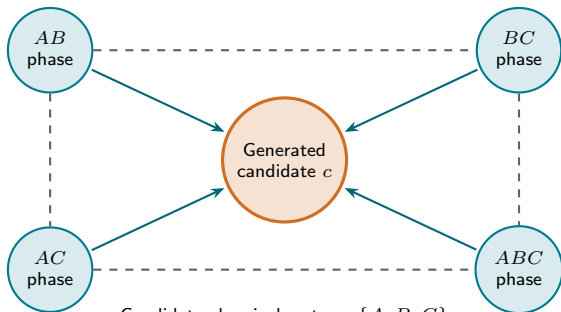
Smaller distance $D(c)$ means a more promising candidate [1].

Limitation

References form an **unstructured, equal-weight set**.

- All retrieved phases contribute equally.
- Reference–reference relationships are discarded.
- The score can be sensitive to redundant or weakly relevant references.

Proposed idea: a local chemistry-aware reference graph



Candidate chemical system: $\{A, B, C\}$
Solid arrows: reference messages to candidate. Dashed lines: reference context.

- **Nodes:** complete relaxed generated crystals and known reference phases.
- **Edges:** first enforce chemical validity; then retain k nearest neighbours in frozen Crys-JEPA embedding space.
- **Edge features:** embedding distance, relation type (exact system / subsystem), and composition similarity.

Model: learn relevance, not merely proximity

For candidate c , construct its allowable reference pool only from training-side phases:

$$R_{\text{valid}}(c) = \{r : \text{elements}(r) \subseteq \text{elements}(c)\}.$$

Build a small local graph with $k \in \{8, 16, 32\}$ retrieved references and apply an edge-aware GNN:

$$\hat{p}_{\text{stable}}(c) = \text{GNN}(G_c).$$

- Start with two-layer GraphSAGE or graph attention.
- Retain a direct candidate-feature path to avoid over-smoothing.
- Predict stability class or a ranking score for low E_{hull} .

Why a graph?

The model can learn that some phase relations are informative while others are not, and can use reference–reference context.

Controlled comparison

Freeze Crys-JEPA embeddings to isolate the effect of graph aggregation under matched supervision.

Experimental protocol: make the claim falsifiable

Data and splits

- Use generated, MatterSim-relaxed candidate structures with a fixed stability-label protocol.
- Build all references from the training split only.
- Hold out related relaxation snapshots together.
- Add formula- or prototype-held-out evaluation for transfer.

Matched baselines

- ① Original Crys-JEPA average-distance score.
- ② Nearest-neighbour and inverse-distance scores.
- ③ Candidate-only MLP on the same frozen embedding.
- ④ Set model seeing the same retrieved references but no graph topology.
- ⑤ Proposed GNN.

Interpretation rule

If the set model matches the GNN, the gain comes from retrieval rather than graph structure.

Evaluation and success criteria

Primary metrics

- Precision@top 10% and top 20% for stable / near-hull candidates.
- Recall of low- E_{hull} candidates.
- AUROC or ranking metrics as secondary measures.
- Screening and graph-construction runtime.

Robustness tests

- Vary k and relation types.
- Remove irrelevant neighbours.
- Duplicate a reference entry: a robust model should not change materially.
- Test transfer to generated candidates and unfamiliar prototypes.

Success

The GNN consistently beats both the original score *and* an equally informed set baseline on held-out and generated candidates.

References

- [1] Nian Liu et al.
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arXiv:2605.14759

- [2] Claudio Zeni et al.
A generative model for inorganic materials design.
Nature, 2025.
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- [3] Han Yang et al.
MatterSim: A Deep Learning Atomistic Model Across Elements, Temperatures and Pressures.
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